PRAXIS

The Multi-Agent Cognitive Simulation & Verifiable Talent Protocol

Flight Simulator for Technical Hiring | Replacing LeetCode with Live Behavioral Sandboxes

CELESTIA 2026 | 24 HR NATIONAL HACKATHON | Open Innovation Track

TEAM NAME: VIPER

MEMBERS:

1. **Sai Vivek K N (Team Lead | Full-Stack & System Architect) - Ph: 7892585383**
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2. **Reehan Pasha (Co-Lead | AI Systems & Infrastructure Engineer) - Ph: 8971183678**
   Email: reehanpasha1607@gmail.com

COLLEGE: Kalpataru Institute of Technology (KIT), Tiptur

01 Problem Statement

What problem are you solving?

THE CRISIS: AI HAS COMPLETELY BROKEN THE TECHNICAL HIRING FUNNEL

> 93% of LeetCode Medium/Hard algorithmic questions and take-home tasks are solved instantly by Claude 3.5 & GPT-4o.
> Remote coding screens suffer from rampant cheating (dual-monitors, background LLMs), creating zero signal for recruiters.
> Whiteboard trivia tests rote memorization under anxiety rather than practical day-to-day software engineering competency.

WHO FACES THIS PAIN?

1. Tech Companies & Startups:
   Spend 42+ Senior Engineer hours per hire ($5,200 salary waste) vetting false positives.
2. Capable Candidates & Self-Taught Devs:
   Filtered out by rigid ATS algorithms due to lack of pedigree degrees or brand-name resumes.

WHY IT MATTERS RIGHT NOW (2026)

> 60%+ surge in AI-generated fake resumes & fraudulent remote interview applications.
> $42,000 average cost of an engineering mis-hire.
> 46% of newly hired developers fail to navigate production codebases in their first 90 days.
> The industry urgently needs: Behavioral Process Observation over static trivia.

02 Objective

What does your solution aim to achieve?

CORE GOAL:

OUTCOME 1: 94% PREDICTIVE ACCURACY

Measure HOW candidates think (debugging discipline, trade-off articulation, keystroke variance) to achieve 94% retention prediction versus 48% with LeetCode.

OUTCOME 2: 70% CYCLE REDUCTION

Compress the entire 4-stage technical loop (screen, take-home, system design, behavioral) into a single sub-45-minute high-signal simulation.

OUTCOME 3: 100% SOVEREIGN VERIFICATION

Issue tamper-proof ZK-SNARK credentials (W3C format) verifying candidate competency without exposing proprietary test scenarios or candidate code.

03 Proposed Solution

Your idea in simple words

THE SOLUTION: THE WORLD'S FIRST PRODUCTION FLIGHT SIMULATOR FOR TALENT

1. PROCEDURAL MUTATION

ProblemBank mutates incident parameters, race conditions & edge cases dynamically so no two tests are identical. Cheating is impossible.

2. MULTI-AGENT PERSONAS

Sub-300ms Groq Llama-3.3-70B agents roleplay as Product Managers (Alex), Architects (Sam), and Clients with hidden business constraints.

3. BEHAVIORAL TELEMETRY

Observer Layer captures keystroke flight-time variance, context switching, structured debugging & cognitive load score (0-100) in real time.

4. ZK TALENT PASSPORT

Generates decentralized SnarkJS Zero-Knowledge Proofs of competency so candidates own their verified score across all employers.

04 Innovation & Uniqueness

What makes it different?

WHY PRAXIS OUTPERFORMS EXISTING APPROACHES

EVALUATION CRITERIA	LEETCODE / HACKERRANK	TAKE-HOME ASSIGNMENTS	PRAXIS (OUR SYSTEM)
Cheating Resistance	0% (Solved by LLM in 5s)	Low (ChatGPT / Ghostwriter)	**100% (Live Behavioral Telemetry)**
Real-World Validity	Low (Abstract algorithms)	Medium (Static code)	**Very High (Incident + AI Team)**
Time Investment	60-90 min stress drill	8-12 hours high fatigue	**< 45 min immersive simulation**
Soft Skills & Pressure	None evaluated	Zero visibility	**Real-time AI stakeholder pushback**
Ownership & Privacy	Locked in company silos	Uncredited discard	**Candidate-Owned ZK-SNARK Passport**

THE UNFAIR ADVANTAGE / MOAT: 'Process Over Answers'. We analyze the micro-second trajectory of how problems are scoped, tested, and resolved under stakeholder tension - something no LLM prompt wrapper can fake.

05 Tech Stack

Tools and technologies you will use

FRONTEND & COCKPIT

> Next.js 16 (App Router) & React 19
> Monaco Browser IDE (VS Code Core)
> XTerm.js (Interactive Cloud Terminal)
> Framer Motion (Visual Telemetry)

AI ORCHESTRATION LAYER

> Groq Llama-3.3-70B Versatile
> Sub-300ms ultra-low latency response
> Multi-Agent Dynamic Personas
> Vercel AI SDK Core Integration

EXECUTION & DATA

> E2B Cloud Sandboxes (MicroVMs)
> Supabase PostgreSQL & Prisma ORM
> Real-time WebSocket Telemetry Bus
> Ephemeral Docker code runners

SECURITY & VERIFICATION

> SnarkJS (circom ZK-SNARK proofs)
> W3C Verifiable Credentials
> Privacy-preserving score attestation
> EEOC-compliant telemetry audit

ENTERPRISE-GRADE ARCHITECTURAL DECISIONS

1. Real-time MicroVMs: Code is executed in isolated, ephemeral Linux microVMs via E2B, completely isolated from our host.
2. Sub-300ms Conversational Latency: Migrated from slow proxies to Groq LPUs, enabling human-speed verbal/chat banter.
3. Zero-Knowledge Proofs: Mathematical proofs guarantee score validity without exposing proprietary company test code.

06 Architecture / Workflow

How the system works

PRAXIS SYSTEM ARCHITECTURE & 6-STEP TALENT LOOP

1. INGESTION
Baseline Litmus test sets initial candidate competency tier.

2. MUTATION
ProblemBank dynamically mutates bugs, variables & SLA targets.

3. COCKPIT
Monaco IDE + E2B microVM terminal + Groq persona chat.

4. OBSERVER
Captures keystroke dynamics, pauses & cognitive load score.

5. EVALUATOR
8-dimensional composite analysis & EEOC narrative report.

6. ZK PASSPORT
SnarkJS attestation seals score to verifiable talent passport.

END-TO-END FLOW: INPUT TO OUTPUT

[CANDIDATE INPUTS] -> [Procedural Incident Bank] -> [E2B Cloud MicroVM Sandbox & Monaco IDE]
|
[BEHAVIORAL SIGNALS] <--- [Live Groq Llama-3.3-70B Personas (Alex-PM, Sam-Architect)] <---|
| (Keystroke timing variance, AST diffs, cognitive load score, ambiguity handling)
v
[AI OBSERVER] -> [Meta-Cognitive Evaluator] -> [SnarkJS ZK-SNARK Attestation] -> [COMPANY DASHBOARD / MARKET]

07 Feasibility

Can it be built in 24 hours?

BUILT IN 24 HOURS: WORKING PROTOTYPE

> Operational Monorepo (Next.js 16 + TypeScript).
> Groq Llama-3.3-70B integrated with sub-300ms speed.
> Monaco IDE with live syntax highlighting & simulation.
> Live behavioral telemetry observer & scoring engine.
> Company & Candidate verification dashboards active.

VIPER TEAM EXECUTION STRENGTHS

> Sai Vivek K N (Lead Architect):
Deep expertise in Next.js, LangGraph orchestration, ZK-SNARK verification & state machines.
> Reehan Pasha (AI Systems Engineer):
Specialist in Groq API optimization, low-latency microVM container pipelines & telemetry heuristics.

RISK MITIGATION & RESILIENCE

> Challenge 1: LLM Latency during roleplay.
Solution: Migrated to Groq LPU engine (<300ms inference).
> Challenge 2: Network & Serverless DB Hangs.
Solution: Built robust client-side bypass & local session caching for zero-friction demo continuity.
> Challenge 3: Sandbox Security.
Solution: E2B cloud microVM isolation.

08 Impact

Who benefits and how

TARGET BENEFICIARIES

> Early-Career & Non-Pedigree Talent:
Get hired based on undeniable proof of process rather than pedigree college stamps.
> Tech Companies & Recruiters:
Eliminate LeetCode cheating, reduce 42 hours of senior engineering interview burn to 45 mins.

BUSINESS & SOCIAL VALUE

> Direct Economic ROI: Saves $3,200+ per screened candidate loop in senior engineer salary.
> Meritocracy: Democratizes hiring across tier-2/3 universities, eliminating implicit human bias.
> Monetization: B2B SaaS ($499/mo per engineering team) + Apprentice Talent Market placement fee.

HORIZONTAL SCALABILITY

> Multi-Domain Ready: Architecture natively supports Engineering, Product Management, Enterprise Sales, and Data Science.
> Cloud-Native: Ephemeral microVMs spin up in <2 seconds with near-zero idle compute cost.
> Global TAM: $1.25B serviceable technical assessment market.

09

Future Scope

Where can this go next?

PHASE 1: REAL-TIME VOICE AGENTS

> Sub-200ms Voice-to-Voice AI stakeholders for spoken phone screens, board meetings, and live client objection roleplay.
> Stress and vocal cadence analysis integrated into the cognitive observer.

PHASE 2: GITHUB INCIDENT INGESTION

> Automatic 60-second scenario generator: connect a company's private GitHub repo and Slack post-mortems.
> System automatically transforms historical production outages into custom mutated sandbox challenges.

PHASE 3: TEAM SIMULATIONS & ATS INTEGRATION

> Multi-candidate synchronous hackathon simulations to assess cross-functional leadership.
> Native 1-click ATS integrations with Greenhouse, Ashby, and Lever.
> Decentralized ZK-SNARK on-chain talent ledger.

10 Team Details

Who is behind the idea

MEMBER 1 (LEAD)

Sai Vivek K N

Team Lead & Architect

> Full-Stack Architecture
> Next.js 16 App Router
> ZK-SNARK Cryptography
> State Machine Design
> UI/UX Cockpit

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MEMBER 2 (CO-LEAD)

Reehan Pasha

AI & Cloud Systems

> Groq Llama-3.3 Pipelines
> Multi-Agent Personas
> E2B MicroVM Sandboxes
> Telemetry Heuristics
> Database & Supabase

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INSTITUTION

Kalpataru Inst. of Tech

Dept. of Computer Science

> Tiptur, Karnataka
> Autonomous Innovation
> Student Developer Cell
> Focused on AI Systems & Decentralized Tech

TEAM PROFILE:

Full 2-Person Core Dev Team
High Execution Velocity

HACKATHON TRACK

CELESTIA 2026

Open Innovation Track

> 24-Hour National Sprint
> Offline Round Qualifier
> Live Functional Prototype
> Enterprise B2B SaaS
> Production Deployment

VERIFICATION:

Live GitHub Monorepo &
Vercel Deployment Verified